

Lecture 09 - 10

Chapter 02: Random Variables

Math 4441: Probability and Statistics

Reference: Goodman & Yates – Introduction to Probability and Stochastic Process, 3rd Edition

Expected Values

Example: Suppose you play a game, where in each play you lose \$1 with probability 0.60 and you win \$1, \$2 and \$3 with probabilities 0.30, 0.08 and 0.02, respectively. Find your net gain in each play of the game.

- If you play only a few times, your gain might depend on your luck
- If you play a large number of times – your luck no longer determines the total gain – rather it will be multiple of average gain

Say, you play the game n times, approximately in $0.6 \times n$ games you will lose \$1 per game and will win \$1, \$2 and \$3 in $0.3 \times n$, $0.08 \times n$ and $0.02 \times n$ games, respectively.

Your total gain is

$$(0.6) \times n \cdot (-1) + (0.3) \times n \cdot (1) + (0.08) \times n \cdot (2) + (0.02) \times n \cdot (3) = (-0.08) \times n.$$

Average loss of \$0.08 per game

The more you play, the less luck interferes and your loss comes to \$0.08 per game

- Let X denote the gain in one play
- Then -0.08 is the expected value of X
- $E[X], \mu_X$, or μ is the expected value of X

x	-1	1	2	3	
$P_X(x)$	0.60	0.30	0.08	0.02	1.00
$x \cdot P_X(x)$	-0.60	0.30	0.16	0.06	-0.08

$$P_X(x) = \begin{cases} 0.60, & x = -1 \\ 0.30, & x = 1 \\ 0.08, & x = 2 \\ 0.02, & x = 3 \\ 0, & o/w \end{cases}$$

$$-1 \times 0.6n + 1 \times 0.3n + 2 \times 0.08n + 3 \times 0.02n = -0.08n$$

$$\text{Or, } -1 \times 0.6 + 1 \times 0.3 + 2 \times 0.08 + 3 \times 0.02 = -0.08$$

$$\text{Or, } -1 \times P_X(-1) + 1 \times P_X(1) + 2 \times P_X(2) + 3 \times P_X(3) = -0.08$$

$$E[X] = \sum_{x \in S_X} x \cdot P_X(x)$$

Definition (Expected Value): The *expected value* of a discrete random variable X with the set of possible values S_X and probability mass function $P_X(x)$ is defined by

$$E[X] = \sum_{x \in S_X} x P_X(x)$$

$E[X]$ exists if this sum converges

The expected value of a random variable X is also called the **mean**, or the **mathematical expectation**, or simply the **expectation** of X . It is denoted by $E[X]$, μ_X , or simply μ .

Example: Three packets are sent from one computer to another. Each packet is successfully sent with a probability of 0.8 irrespective of the others. Each successfully sent packet results in a reward of 1 unit whereas 1 unit penalty (negative reward) is imposed if none of the packets is sent successfully. Find the expected net reward.

Let X is the net reward. X is a discrete random variable with values

$$S_X = \{-1, 1, 2, 3\}$$

$$P[X = -1] = 0.008$$

$$P[X = 1] = 0.096$$

$$P[X = 2] = 0.384$$

$$P[X = 3] = 0.512$$

$$E[X] = -1 \times 0.008 + 1 \times 0.096 + 2 \times 0.384 + 3 \times 0.512$$

Example: A college mathematics department sends 8 to 12 professors to the annual meeting of the American Mathematical Society, which lasts five days. The hotel at which the conference is held offers a bargain rate of a dollars per day per person if reservations are made 45 or more days in advance, but charges a cancellation fee of $2a$ dollars per person. The department is not certain how many professors will go. However, from past experience it is known that the probability of the attendance of i professors is $1/5$ for $i = 8, 9, 10, 11$, and 12 . If the regular rate of the hotel is $2a$ dollars per day per person, should the department make any reservations? If so, how many?

$$E(X_8) = (40a)\frac{1}{5} + (50a)\frac{1}{5} + (60a)\frac{1}{5} + (70a)\frac{1}{5} + (80a)\frac{1}{5} = 60a$$

$$E(X_9) = (42a)\frac{1}{5} + (45a)\frac{1}{5} + (55a)\frac{1}{5} + (65a)\frac{1}{5} + (75a)\frac{1}{5} = 56.4a,$$

$$E(X_{10}) = (44a)\frac{1}{5} + (47a)\frac{1}{5} + (50a)\frac{1}{5} + (60a)\frac{1}{5} + (70a)\frac{1}{5} = 54.2a,$$

$$E(X_{11}) = (46a)\frac{1}{5} + (49a)\frac{1}{5} + (52a)\frac{1}{5} + (55a)\frac{1}{5} + (65a)\frac{1}{5} = 53.4a,$$

$$E(X_{12}) = (48a)\frac{1}{5} + (51a)\frac{1}{5} + (54a)\frac{1}{5} + (57a)\frac{1}{5} + (60a)\frac{1}{5} = 54a.$$

X_{11} has the smallest expected value. Therefore, making 11 reservation is the most reasonable policy

Derived Random Variable:

If x is a sample value of a random variable X , and y is a sample value of a random variable Y . If we have a function $Y = g(X)$ and we get the values of Y from the values of X . Because we obtain Y from another random variable, we refer to Y as a **derived random variable**.

Definition 2.4 (Derived Random Variable). If X and Y are two random variables, and each sample value y of Y is a mathematical function $g(x)$ of X , then the relationship of these two random variables is given by

$$Y = g(X),$$

and the random variable Y is called a derived random variable.

What will be the PMF of Y ?

For a discrete random variable Y , if there exists another random variable X , where $Y = g(X)$, then the PMF of X is

$$P_Y(y) = \sum_{x:g(x)=y} P_X(x)$$

Expected Value of a derived random variable

$$E[X + Y] = ?$$

$$E[aX + b] = ?$$

n -th Moment of a Random Variable

Variance: Variance measures the average magnitude of the fluctuations of a random variable from its expectations.

$$E[X - E[X]] = E[X - \mu] = E[X] - \mu = E[X] - E[X] = 0.$$

Hence, $E[X - E[X]]$ is not an appropriate measure for the variance.

if we consider $E(|X - E[X]|)$ instead, the problem of negative and positive deviations canceling each other disappears.

mathematically, $E(|X - E[X]|)$ is difficult to handle

The quantity $E[(X - E[X])^2]$ is used to measure the average amount of fluctuations, and is called the variance of X

The square root of $E[(X - E[X])^2]$ is called the **standard deviation** of X .

Definition 2.7. Let X be a discrete random variable with a set of possible values S_X , probability mass function $P_X(x)$, and $E(X) = \mu$. Then σ_X and $Var[X]$, called the **standard deviation** and the **variance** of X , respectively, and are defined by

$$\sigma_X = \sqrt{E[(X - \mu)^2]},$$

and

$$Var[X] = E[(X - \mu)^2].$$

Note that by this definition, we have

$$Var[X] = E[(X - \mu)^2] = \sum_{x \in S_X} (x - \mu)^2 P_X(x).$$

$$\text{Var}[X] = E[X^2] - (E[X])^2.$$

Proof.

$$\begin{aligned}\text{Var}[X] &= E[(X - \mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 \\ &= E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2 \\ &= E[X^2] - (E[X])^2.\end{aligned}$$

$Var[X] = 0$ if and only if X is a constant with probability 1.

$$Var[aX + b] = a^2 Var[X]$$

$$P_X(x) = \begin{cases} 0.60, & x = -1 \\ 0.30, & x = 1 \\ 0.08, & x = 2 \\ 0.02, & x = 3 \\ 0, & o/w \end{cases}$$

$$E[X] = \mu = 0.08$$

$$V[X] = [-1 - (-0.08) \times 0.60 + (1 - 0.08) \times 0.30 + (2 - 0.08) \times 0.08 + (3 - .08) \times 0.02$$

n-th Central Moment

Expected Value and Variance for Continuous Random Variables

$$E[X] = \int_{-\infty}^{+\infty} x \cdot f_X(x) dx$$

$$V[X] = \int_{-\infty}^{+\infty} (x - \mu)^2 \cdot f_X(x) dx$$

Example: The time elapsed, in minutes, between the placement of an order of pizza and its delivery is random with pdf

$$f(x) = \begin{cases} 1/15 & \text{if } 25 < x < 40 \\ 0 & \text{otherwise.} \end{cases}$$

$$E(X) = \int_{25}^{40} x \frac{1}{15} dx = 32.5,$$

$$E(X^2) = \int_{25}^{40} x^2 \frac{1}{15} dx = 1075.$$

$$V[X] = 1075 - (32.5)^2 = 18.75$$

$$\sigma_X = \sqrt{V[X]} = 4.33$$

Sketch the graph of the function

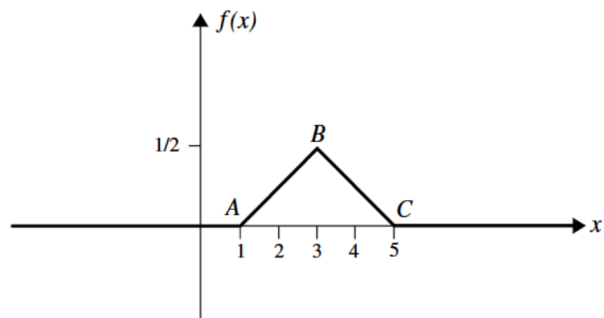
$$f(x) = \begin{cases} \frac{1}{2} - \frac{1}{4}|x - 3| & 1 \leq x \leq 5 \\ 0 & \text{otherwise,} \end{cases}$$

and show that it is the probability density function of a random variable X .

Find F , the distribution function of X , and show that it is continuous.

Find $E[X]$ and $V[X]$.

$$f(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{2} + \frac{1}{4}(x - 3) & 1 \leq x < 3 \\ \frac{1}{2} - \frac{1}{4}(x - 3) & 3 \leq x < 5 \\ 0 & x \geq 5. \end{cases}$$



For $t < 1$,

$$F(t) = \int_{-\infty}^t 0 \, dx = 0;$$

for $1 \leq t < 3$,

$$F(t) = \int_{-\infty}^t f(x) \, dx = \int_1^t \left[\frac{1}{2} + \frac{1}{4}(x-3) \right] dx = \frac{1}{8}t^2 - \frac{1}{4}t + \frac{1}{8};$$

for $3 \leq t < 5$,

$$\begin{aligned} F(t) &= \int_{-\infty}^t f(x) \, dx = \int_1^3 \left[\frac{1}{2} + \frac{1}{4}(x-3) \right] dx + \int_3^t \left[\frac{1}{2} - \frac{1}{4}(x-3) \right] dx \\ &= \frac{1}{2} + \left(-\frac{1}{8}t^2 + \frac{5}{4}t - \frac{21}{8} \right) = -\frac{1}{8}t^2 + \frac{5}{4}t - \frac{17}{8}; \end{aligned}$$

and for $t \geq 5$,

$$\begin{aligned} F(t) &= \int_{-\infty}^t f(x) \, dx = \int_1^3 \left[\frac{1}{2} + \frac{1}{4}(x-3) \right] dx + \int_3^5 \left[\frac{1}{2} - \frac{1}{4}(x-3) \right] dx \\ &= \frac{1}{2} + \frac{1}{2} = 1. \end{aligned}$$